

$$(1) \pi_{SNO}(\sigma_{UNO='U1'}(SPJ))$$

$$(2) \pi_{SNO}(\sigma_{UNO='U1', APNO='P1'}(SPJ))$$

$$(3) \pi_{SNO}(\sigma_{COLOR='红'}(SPJ \bowtie P))$$

$$(4) \pi_{UNO}(J) - \pi_{UNO}(\sigma_{CITY='天津'}(S) \bowtie SPJ \bowtie \sigma_{COLOR='红'}(P))$$